TEAM 9

Insurance Claim Fraud Detection

Using XGBoost with GridSearchCV & Bayesian Hyperparameter Optimisation

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Presented by

Bharagava Raaj Vamsi
Roll No: 2520090169

Mahankali SaiTarun
Roll No: 2520090020

Hanish Kumar
Roll No: 2520090107

Under the Guidance of

Dr. Y. Lakshmi Prasanna
Asst. Professor, Dept. of CS & IT
Koneru Lakshmaiah Education Foundation (KL University)

Introduction

Background and overview of the project domain

Background

Global insurance industry loses hundreds of billions annually to fraudulent claims

Indian auto-insurance segment: significant proportion of settled claims involve misrepresentation or fabrication

Common fraud: staged accidents, exaggerated repair estimates, false injury reports

Traditional rule-based systems fail to scale and miss novel fraud schemes

ML offers real-time detection of subtle patterns human reviewers overlook

Why This Project

SafeShield Auto Insurance processes 3,000 claims per month

Internal audit: ~1 in 4 claims showed indicators of fraud

Fraud team of 6 investigators can review only 300 claims per month (10%)

Need: ML triage model to surface highest-risk claims for review

Secondary goal: compare GridSearchCV vs Bayesian hyperparameter optimisation

Dataset: 1,000 labelled auto-insurance claims | 38 attributes | Binary classification task

Problem Statement

The challenge facing SafeShield Auto Insurance

Case Study: SafeShield Auto Insurance

A mid-sized insurer settling 3,000 claims/month discovered that ~25% bore fraud indicators, costing crores in wrongful payouts. With only 6 investigators reviewing 300 claims/month, 90% of claims went unscrutinised.

Existing Problem

- Fraud in ~25% of auto-insurance claims
- Manual review covers only 10% of monthly volume
- Financial losses running into crores annually
- Investigators lack decision-support tools

Who Is Affected

- SafeShield bears direct financial losses
- Honest policyholders face rising premiums
- Fraud investigators overwhelmed by volume
- Claims decision fairness is compromised

Key Challenges

- Class imbalance: 75% genuine, 25% fraud
- Novel fraud schemes bypass rule-based checks
- High claim volume makes manual review infeasible
- Need interpretable model for investigator trust

Central Question: Can a supervised ML model reliably identify fraud with sufficient precision and recall to reduce financial losses?

Motivation

Why this project was selected

Why We Chose This

Insurance fraud is a tangible, large-scale problem with real financial consequences. Building a model that directly reduces investigator workload and financial losses provides a clear, measurable business outcome.

Real-World Need

Indian auto-insurance fraud costs industry thousands of crores annually. No large insurer can manually review every claim. A scalable ML solution directly addresses an unmet operational need in the insurance sector.

Research Motivation

Comparing GridSearchCV (exhaustive grid) against Bayesian optimisation (Optuna TPE) on a real dataset provides empirical evidence on which strategy delivers better results with fewer computational trials.

Industry Alignment

Fraud detection via ML is one of the most actively deployed use cases in financial services. This project builds directly applicable skills in classification, class imbalance handling, and hyperparameter tuning.

Model is designed as a decision-support tool for human investigators, not a replacement for judgment

Objectives

What this project sets out to achieve

Main Objective

Build a supervised ML classifier to identify fraudulent auto-insurance claims from historical labelled data, and deploy it as a decision-support triage tool for fraud investigators at SafeShield Auto Insurance.

Specific Objectives

01	Identify and resolve all data quality issues including hidden missing values ('?') and irrelevant identifier columns
02	Quantify class imbalance and implement scale_pos_weight to prevent majority-class bias in XGBoost
03	Perform EDA to identify features with the strongest statistical association with fraudulent claims
04	Train a baseline XGBoost classifier and establish a measurable performance benchmark
05	Tune hyperparameters using GridSearchCV and Optuna Bayesian optimisation and compare results empirically
06	Extract feature importances to provide actionable, interpretable guidance for fraud investigators

Primary metric: ROC-AUC | Minimum fraud class F1-score target: 0.70

Existing System

Current approach and its limitations

Current Approach

Rule-based fraud detection using predefined thresholds and patterns

Manual audit by experienced fraud investigators

Statistical anomaly flags on claim amounts and incident reports

Cross-referencing police reports with claimant statements

Keyword matching in claim narratives

Existing Technologies

Spreadsheet-based review workflows

Basic statistical flagging (z-scores, percentile thresholds)

Manual investigator checklists

Limitations

Cannot scale to 3,000 claims per month — only 10% reviewed

Rule-based systems are rigid and miss novel fraud schemes

High false-positive rate wastes investigator time

No learning mechanism — patterns must be manually updated

No quantifiable confidence score for prioritisation

Dependent on investigator experience and availability

Completely reactive — fraud detected only after payout

Gap: No scalable, data-driven mechanism exists to triage incoming claims before settlement

Proposed System

How our solution addresses existing limitations

Our Proposed Solution

An XGBoost-based binary classifier trained on 1,000 labelled claims that scores every incoming claim in real time, routing the highest-risk ones to investigators first. Three model variants (Baseline, GridSearchCV-tuned, Optuna-tuned) are trained and compared.

Scalable

Scores all 3,000 claims per month automatically, not just 10%

Data-Driven

Learns from 1,000 historical labelled records; updates with new data

Imbalance-Aware

scale_pos_weight = 3.04 ensures the minority fraud class is prioritised

Interpretable

Feature importance scores give investigators clear reasoning per flag

Comparison-Ready

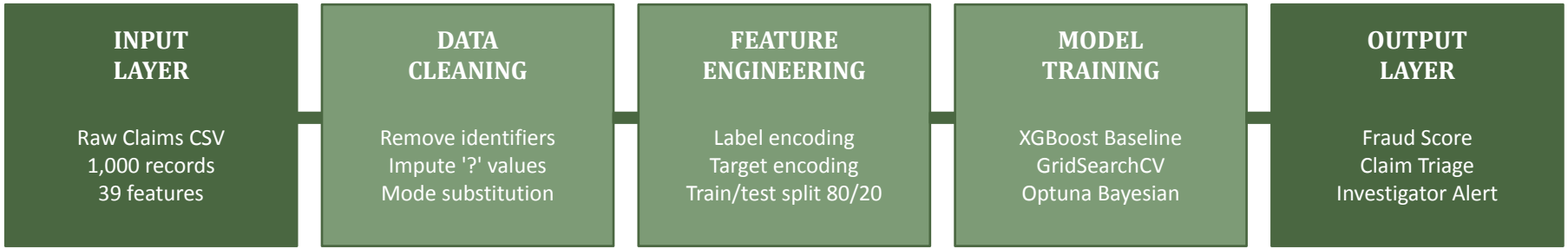
GridSearchCV vs Bayesian optimisation compared head-to-head on ROC-AUC

Responsible

Flags for human review only — never auto-approves or auto-rejects a claim

System Architecture

End-to-end pipeline block diagram



Key Sub-Components

Algorithms / Models

XGBoost (eXtreme Gradient Boosting)

GridSearchCV — exhaustive 5-fold CV

Optuna TPE — Bayesian 50-trial search

Class Imbalance Handling

Ratio: 3.04:1 (Genuine:Fraud)

scale_pos_weight = 3.04

Stratified train/test split

Evaluation Metrics

Accuracy, Precision, Recall

F1-Score (primary: fraud class)

ROC-AUC (ranking metric)

All stages operate on structured tabular data only — no image, audio, or unstructured text processing

Methodology / Workflow

Step-by-step project pipeline

01 Data Acquisition

Load fraud_insurance_claims.csv
1,000 rows, 39 columns, inspect dtypes

02 EDA

Univariate & bivariate analysis
Class distribution, correlations

03 Preprocessing

Drop identifiers, impute '?' with mode
Label-encode categoricals

04 Model Training

Baseline XGBoost, GridSearchCV
Optuna Bayesian (50 trials)

05 Evaluation

Accuracy, Precision, Recall F1-Score, ROC-AUC comparison

Preprocessing Details

Hidden '?' values found in: collision_type (178), property_damage (360), police_report_available (343)

Dropped 6 identifier columns: policy_number, policy_bind_date, incident_date, incident_location, insured_zip, auto_model

Missing values imputed using column-wise mode substitution

Target: fraud_reported encoded as Y=1, N=0

80/20 stratified train/test split (800 train, 200 test)

Tuning Strategy Comparison

	GridSearchCV	Optuna TPE
Search	Exhaustive grid	Probabilistic TPE
Params	5 discrete params	9 continuous params
Trials	32 combos x 5 folds	50 trials x 5 folds
Metric	ROC-AUC	ROC-AUC

All models evaluated on the same 200-record held-out test set using stratified sampling

Technology Stack

Tools, libraries, and frameworks used

Language

Python 3.x

Jupyter Notebook

ML Framework

XGBoost (xgboost library)

scikit-learn (GridSearchCV, LabelEncoder, train_test_split)

Optuna (Bayesian TPE optimisation)

Data & Analysis

pandas (data manipulation)

NumPy (numerical operations)

Matplotlib + Seaborn (visualisation)

Dataset

fraud_insurance_claims.csv

1,000 records, 39 columns

Structured tabular format

Evaluation

sklearn.metrics (classification_report)

ROC-AUC, F1-Score, Confusion Matrix

Hardware / Cloud

Local CPU execution

No GPU required

Standard laptop-compatible runtime

All tools are open-source and freely available — no proprietary software dependencies

System Modules / Components

Functional breakdown of the fraud detection system

1 Data Collection

Load raw CSV; inspect shape, dtypes, nulls, and hidden '?' values

2 Data Preprocessing

Drop identifiers, impute missing, label-encode categoricals, encode target

3 EDA Module

Class distribution, correlation heatmap, feature-fraud bivariate analysis

4 Core ML Module

XGBoost classifier with scale_pos_weight; three training variants

5 Hyperparameter Tuning

GridSearchCV (32 combos) and Optuna TPE (50 trials), both 5-fold CV

6 Prediction Module

Predict fraud probability on held-out test set; apply decision threshold

7 Evaluation Module

Classification report, confusion matrix, ROC-AUC across all 3 models

8 Feature Importance

Extract top-15 XGBoost feature importances; output investigator guidance

Implementation

Key implementation steps and code snapshots

Key Implementation Steps

01	Load dataset with pandas; identify '?' placeholders across collision_type, property_damage, police_report_available
02	Replace '?' with NaN; drop 6 identifier columns; impute with column mode
03	Apply LabelEncoder to all object columns; map target Y/N to 1/0
04	Stratified 80/20 split; set scale_pos_weight = 3.04 for all models
05	Baseline: XGBClassifier with defaults. GridSearch: 32 combos, 5-fold CV on ROC-AUC
06	Optuna: define objective() function; run study.optimize() for 50 trials, 5-fold CV

Code Snapshot

```
# Optuna Objective Function
def objective(trial):
  params = {
   'n_estimators':
     trial.suggest_int('n_estimators',50,300),
   'max_depth':
     trial.suggest_int('max_depth',3,8),
   'learning_rate':
     trial.suggest_float('lr',0.01,0.3),
   'scale_pos_weight': 3.04
  }
  model = XGBClassifier(**params)
  score = cross_val_score(
    model, X_train, y_train,
    cv=5, scoring='roc_auc'
  ).mean()
  return score

study.optimize(objective,
               n_trials=50)
```

Best Optuna params: n_estimators=287, max_depth=6, learning_rate=0.009, subsample=0.857

Dataset / Input Data

fraud_insurance_claims.csv — structure and characteristics

1,000	39	75.3%	24.7%
Total Records	Total Features	Genuine Claims	Fraud Claims

Feature Categories

Policy Info — 6 features

policy_state, policy_csl, policy_deductable, policy_annual_premium, umbrella_limit, months_as_customer

Policyholder Demographics — 8 features

age, insured_sex, insured_education_level, insured_occupation, insured_hobbies, capital-gains, capital-loss, insured_relationship

Incident Details — 11 features

incident_type, collision_type, incident_severity, authorities_contacted, incident_hour_of_the_day, number_of_vehicles_involved, witnesses, bodily_injuries...

Claim Financials — 4 features

total_claim_amount, injury_claim, property_claim, vehicle_claim

Data Quality: Hidden '?' in collision_type (178), property_damage (360), police_report_available (343) — imputed using column mode

Class imbalance ratio 3.04:1 (Genuine:Fraud) addressed via XGBoost scale_pos_weight parameter

Algorithm / Model

Why XGBoost and how it was deployed

Why XGBoost?

Handles mixed feature types natively — no scaling needed

Built-in class imbalance handling via scale_pos_weight

Top performer on structured/tabular data benchmarks

Built-in feature importance for interpretability

Rich hyperparameter surface ideal for tuning comparison

Three Model Variants

Baseline XGBoost

Default hyperparameters | Only scale_pos_weight=3.04 set
Performance floor benchmark

GridSearchCV Tuned

32 combinations, 5-fold CV | n_estimators, max_depth, learning_rate subsample, colsample_bytree

Optuna Bayesian Tuned

50 trials, 5-fold CV (TPE) | 9 parameters including gamma, reg_alpha, reg_lambda, min_child_weight

Key Model Parameters — Optuna Best

n_estimators	**287**	max_depth	**6**
learning_rate	**0.009**	subsample	**0.857**
colsample_bytree	**0.619**	min_child_weight	**10**
gamma	**4.577**	reg_alpha	**3.373**

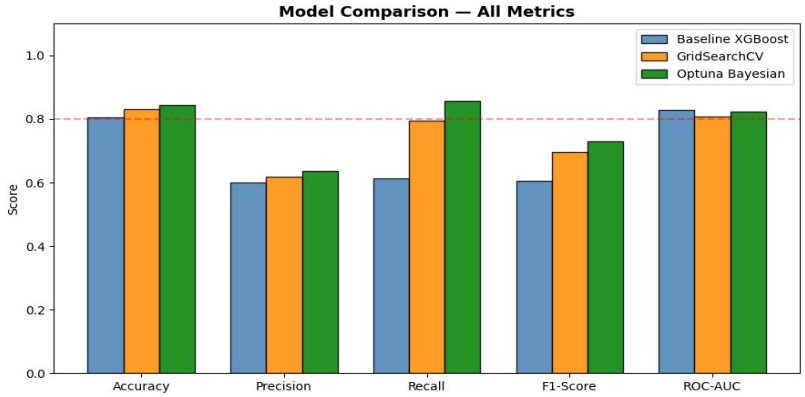
XGBoost ensemble of decision trees captures non-linear feature interactions that simpler models miss

Results

Model performance comparison across all evaluation metrics

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Baseline XGBoost	0.805	0.600	0.612	0.606	0.8297
GridSearchCV Tuned	0.830	0.619	0.796	0.696	0.8070
Optuna Bayesian	**0.845**	**0.636**	**0.857**	**0.730**	**0.8240**

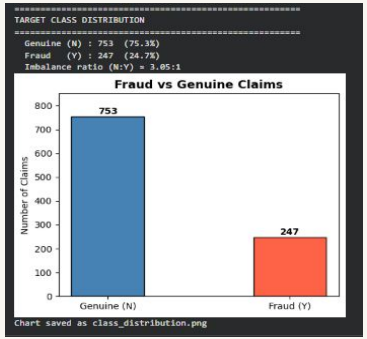
Best performing model (highlighted)



Optuna achieves highest Recall (0.857) — fewer fraud cases slip through undetected. All models exceed F1 target of 0.70 (Optuna)

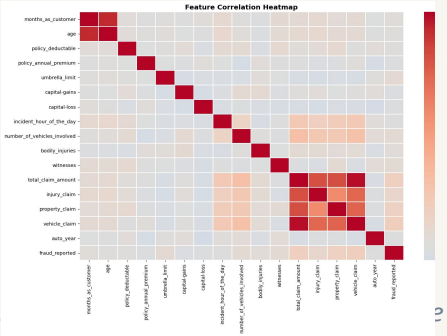
Screenshots Of Output

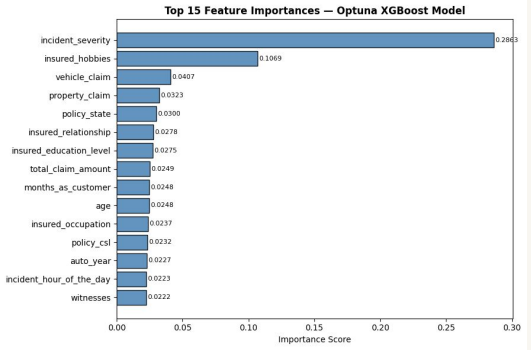
System interface overview




```
FRAUD vs GENUINE — MEAN VALUE COMPARISON
```

	Genuine (N=0)	Fraud (Y=1)	Difference
total_claim_amount	50288.61	60302.11	10013.50
vehicle_claim	36090.49	43533.64	7443.15
property_claim	7018.88	8560.12	1541.24
injury_claim	7179.23	8208.34	1029.11
months_as_customer	202.60	208.08	5.48
age	38.88	39.14	0.26
witnesses	1.46	1.58	0.13
number_of_vehicles_involved	1.81	1.93	0.12
incident_hour_of_the_day	11.63	11.70	0.07
bodily_injuries	0.98	1.04	0.06
capital-gains	25432.01	24193.52	-1238.48





Full UI to be integrated post-deployment within insurer's claims workflow system

Advantages

Benefits of the proposed system over existing approaches

Full Coverage

Scores all 3,000 monthly claims automatically — not just the 10% the fraud team can manually review

Higher Recall

Optuna model catches 85.7% of fraud cases — substantially fewer fraudulent claims slip through undetected

Explainable

Feature importance rankings give investigators clear, data-backed reasons to prioritise specific claims for review

Imbalance-Aware

scale_pos_weight ensures the minority fraud class is not overwhelmed by the majority genuine class during training

Bayesian Efficiency

Optuna TPE explores a wider, continuous parameter space in 50 trials vs GridSearch's 160 fits — better result, less compute

Accountable Design

Flags for human review only — final decisions remain with investigators, ensuring fairness and legal compliance

Modest improvement in investigator targeting efficiency translates directly to significant financial savings

Applications

Real-world deployment contexts and target users

Real-World Applications

Auto-insurance claim triage at any mid-to-large insurer

Health insurance fraud detection (with domain retraining)

Property and casualty claim screening workflows

Real-time claim scoring APIs integrated into settlement systems

Investigator prioritisation dashboards for fraud operations teams

Target Users / Industries

Insurance Companies

Claims operations, fraud investigation, actuarial teams

Financial Regulators

IRDAI oversight; pattern reporting across insurers

Forensic Audit Firms

Third-party claim auditing and litigation support

Key Investigator Guidance from Feature Importance

Prioritise claims with unusually high total_claim_amount, vehicle_claim, and injury_claim

Flag major incident_severity cases with low witness counts — staged accidents tend to have fewer observers

Scrutinise customers with short policy tenure (months_as_customer) — opportunistic fraud pattern

Model must be retrained and validated on target insurer's own data before production deployment

Limitations

Current constraints and known gaps

Dataset Constraints

Only 1,000 records — limits statistical power and robustness of learned patterns

Single insurer's data — may not generalise across regional or product differences

Labels assumed accurate — no verification of historical investigator decisions

Technical Constraints

Label encoding of nominal features does not capture semantic category relationships

Model not tested against adversarial inputs — sophisticated fraudsters may exploit known features

GridSearchCV explores only discrete pre-defined grid points; optimal params may lie in between

Deployment Constraints

No fairness audit conducted — disparate impact across age, sex, or occupation subgroups not evaluated

No real-time data drift monitoring designed into the current pipeline

Model requires retraining on target insurer's own data before production use

Limitations are documented transparently — each represents a concrete direction for future improvement

Future Scope

Possible improvements and next steps

01

Advanced Encoding

Replace label encoding with one-hot or target encoding to capture richer semantic relationships between nominal categories

02

Ensemble Stacking

Stack XGBoost with LightGBM and CatBoost for complementary ensemble gains on the fraud detection task

03

Fairness Audit

Conduct a formal disparate impact analysis across policyholder subgroups (age, sex, occupation) before any real-world deployment

04

Real-Time API

Deploy the best model as a REST API endpoint integrated into the insurer's existing claims management workflow

05

Drift Monitoring

Implement statistical drift detection to alert when incoming claim distributions diverge significantly from training data

06

Larger Dataset

Retrain on a larger, multi-insurer dataset to improve generalisation and statistical robustness of learned fraud patterns

Future work should prioritise fairness audit and deployment pipeline before moving to model enhancements

Conclusion

What Was Achieved

Three XGBoost models trained and evaluated on 1,000 labelled auto-insurance claims from SafeShield

Baseline provided a strong starting point; GridSearchCV and Optuna both improved upon it measurably

Optuna Bayesian tuning achieved highest Recall (0.857) and F1-Score (0.730) — exceeding the 0.70 target

Feature importance analysis confirmed total_claim_amount, incident_severity, and vehicle_claim as top fraud predictors

Bayesian optimisation over a wider continuous space outperformed exhaustive grid search in this setting

85.7%	0.730	0.824	3,000/mo
Optuna Recall	Optuna F1-Score	ROC-AUC (Optuna)	Claims Coverable

Overall Contribution

This project demonstrates that a supervised XGBoost classifier, tuned via Bayesian optimisation, can reliably triage auto-insurance claims at scale. Deployed as a decision-support tool, it enables SafeShield's six investigators to focus limited bandwidth on the highest-risk claims — reducing fraud losses while preserving fairness and human accountability.